Triple Ergodic Averages:
Ram Oscillation Certificates, Local Improvements, and Conditional Extensions

Technical Report M295

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Abstract

We examine the October 4, 2026 manuscript claiming all-integer pointwise convergence of the averages $N^{-1}\sum_{n=1}^{N} f_1(T^n x)f_2(T^{2n}x)f_3(T^{3n}x)$ for an invertible mixing transformation on an arbitrary probability space. Its finite oscillation assertion is stated for every invertible probability-preserving system, a stronger analytic scope than its headline. The private Ram report TR-item-168 addresses this exact operator and supplies directly relevant deterministic oscillation, transference, and approximation principles, but does not itself prove the missing uniform analytic estimate.

We give complete proofs of a robust oscillation criterion, an explicit three-moment correction with asymptotically optimal total variation, square-root delay and logarithmic-3/2 row refinements, and a second-order lattice-planting error. Conditional on the manuscript's uniform analytic closure, we prove the weak-mixing version and the $L^{p_1} \times L^{p_2} \times L^{p_3}$ extension through $\sum_i 1/p_i = 1$. The same analytic premise, combined with established norm convergence, would imply bounded triple pointwise convergence for all invertible systems. These are conditional consequences, not an independent resolution of that problem. Finite physical-Gram and leakage certificates are separated from infinite-depth claims. All seven source sections were read; the much larger companion Hilbert-transform proof was consulted selectively, and its global closure is not certified here.

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1 Statement, status, and applicability

1.1 The exact target

Let $(X, \mathcal{F}, \mu)$ be a probability space, with no standardness assumption, and let T be invertible, bimeasurable, and probability-preserving. Put $Uf = f \circ T$ and

$$A_N(\boldsymbol{f})(x) = \frac{1}{N} \sum_{n=1}^{N} f_1(T^n x) f_2(T^{2n} x) f_3(T^{3n} x). \tag{1.1}$$

The source [1], Introduction, Theorem `thm:main`, assumes ordinary mixing:

$$\mu(T^{-n} A \cap B) \longrightarrow \mu(A)\mu(B) \quad (A, B \in \mathcal{F}), \tag{1.2}$$

and claims, for each bounded complex triple,

$$A_N(\boldsymbol{f})(x) \longrightarrow p(\boldsymbol{f}) := \prod_{i=1}^{3} \int f_i \, d\mu \quad \text{for almost every } x, \qquad N \to \infty \text{ through all positive integers.} \tag{1.3}$$

The null set may depend on the triple. This is not merely convergence along powers of two, convergence in L^2, or convergence of the integrated correlations. There is no asserted common pointwise rate, no universal exceptional set for uncountably many triples, and no uniformity over translated averaging windows.

The pinned source is the repository state `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. It is treated as a new public mathematical claim, not established independent truth. Its companion [2], dated October 5, is another new claim. Neither the present reading nor finite computational tests constitute a certification of their full analytic proofs.

1.2 Primary literature as of October 7, 2026

The following distinctions are essential.

- Host and Kra prove L^2 convergence of bounded arithmetic-progression averages for invertible probability-preserving systems [5, Theorem 1.1]. This identifies a norm limit, not pointwise convergence.
- Bourgain's double-recurrence theorem is bilinear [6]. Krause, Mirek, and Tao treat the bilinear pair $n, P(n)$, with $\deg P \geq 2$ [7]. Neither is a theorem for the triple $n, 2n, 3n$.
- Huang, Shao, and Ye prove the arithmetic-progression pointwise theorem for ergodic distal systems [8]. Gutman, Huang, Shao, and Ye prove product-mean pointwise convergence for weakly mixing pairwise independently determined systems [9, Main Theorem; Theorem 3.4]. The PID condition is additional, not a consequence of ordinary mixing supplied by these results.
- The September 2026 version of Kosz, Mirek, Peluse, Wan, and Wright proves a multilinear theorem for distinct polynomial degrees [10, Theorems 1.3 and 1.12]. In Section 1.8, item 5, it explicitly identifies $n, 2n, 3n$ as an open equal-degree problem beyond its method. Replacing the powers by the commuting maps T, T^2, T^3 does not make the degrees distinct.

The November 2025 progress page of Frantzikinakis also distinguishes the distal triple result from the solved bilinear polynomial problem [11, Problem 11]. These dated primary sources establish the earlier status and scope. They do not rule on the correctness of the later October manuscript. No independent acceptance or full verification of the October analytic argument is asserted here.

The bibliography uses the publisher-confirmed DOI `10.1007/s11854-019-0061-3` for [8]; the different DOI printed in the consulted TR-item-168 bibliography is not used. All public searches concerned public titles or metadata. The confidential corpus was read locally only.

1.3 Why the Ram match is direct

TR-item-168 [3], equations (1.1), (3.1), and (5.1), studies exactly (1.1), including its equal-degree resonant plane, pointwise obstruction, and discrete-to-ergodic passage. This is **Level 5 applicability**: the core problem and operator coincide. It is not a statement that the corpus already proves the source's central theorem.

Local source	Concrete use	Boundary
TR-item-168, deterministic oscillation and skeleton sections	Fixed endpoint selection, all-integer passage, approximation closure	No general trilinear oscillation estimate is supplied
TR-item-168, finite-horizon section	Orbit error budgets and nonlinear product consistency	No infinite-horizon rate from a finite model
TR248, Sections 2–4 and moment assembly	Physical Gram matrices, positive leakage, verified finite margins	Finite-dimensional certification is not continuum Bessel closure
M295, Section 4	Persistent columns and common-input Bessel rows	Whole-cell chronology and all row hypotheses must remain intact

2 The source proof and its global dependencies

2.1 The oscillation premise is independent of mixing

An admissible averaging profile is a real, compactly supported smooth function φ of mass one with

$$\int t^j \varphi(t)\,dt = 0 \quad (1 \le j \le 3), \qquad \left\|\varphi^{(n)}\right\|_\infty \le [G(n+2)]^{G(n+2)} \tag{2.1}$$

for some finite profile-dependent G. It need not be nonnegative. For one-bounded inputs put

$$F_n = \prod_{j=1}^{3} U^{jn} f_j, \qquad V_k^\varphi = \sum_{n\in\mathbb{Z}} 2^{-k}\varphi(n/2^k)F_n. \tag{2.2}$$

The source's Proposition `prop:orbit-oscillation`, in `sections/ergodic.tex`, claims

$$\sum_{i=1}^{w} \int \max_{k_i\le \ell\le k_{i+1}} |V_\ell^\varphi - V_{k_i}^\varphi|\,d\mu \le C_\varphi\sqrt{w} \tag{2.3}$$

for every deterministic increasing sequence $0 \le k_1 < \cdots < k_{w+1}$, every invertible probability-preserving T, and every such triple. The constant may depend on φ, but not on w, the endpoints, the system, or its inputs. Call the universal assertion (2.3) **premise** O. Whenever a conclusion below invokes O, its analytic verification is a substantive condition.

The source proves O from a depth-varying local form theorem, not by applying mixing at an analytic step. For a dyadic cell I, $r = |I|$, its form is

$$H_I(z_0,z_1,z_2,z_3) = r^{-2}\int_{\mathbb{R}^2} w_I(x,t)\prod_{j=0}^{3} z_j(x+jt)\,dx\,dt. \tag{2.4}$$

All four coordinates lie a fixed positive fraction of r inside I. Each one-coordinate line marginal of w_I vanishes; the derivative bounds are of the fixed Gevrey type. For each slot a deterministic partition into W_j consecutive depth blocks satisfies, pointwise on each block,

$$\sup_s |f_{j,s}| + \sum_{\substack{s,s+1 \\ \text{inside the block}}} |f_{j,s+1} - f_{j,s}| \le W_j^{-1/2}. \tag{2.5}$$

The claimed estimate is

$$\sum_{I \subset J} |I|\,|H_I(f_{0,s(I)}, f_{1,s(I)}, f_{2,s(I)}, f_{3,s(I)})| \le C_{\mathrm{loc}}|J|, \tag{2.6}$$

uniformly in all finite original depth trees and all four schedules. Missing depths are still assigned input values. A restart does not reset an input's original depth.

2.2 What each section contributes

The complete seven-section source was read, including its two chronology figures and bibliography. The following description records the actual logical dependencies rather than inferring them from section titles.

Section 2: ergodic conclusion. The deterministic-block argument turns (2.3) into almost-everywhere convergence of each fixed smooth profile. A fixed-lag van der Corput induction identifies the norm limit under mixing. Signed moment corrections approximate positive interval averages. A countable set of dilated dyadic skeletons and finite rational grids then give all N. The order of limits is important at every one of these steps.

Section 3: transference. A fourth-primitive construction produces all four exact line cancellations in (2.4). Averaging translations of a single nested lattice reconstructs a smooth scale difference. Thin neighborhoods of integers transfer continuous estimates to the integer progression. The maximizing endpoint and its complex phase are linearized into a depth schedule satisfying (2.5). An orbit segment is padded through the largest *fixed* endpoint; only then is its spatial length sent to infinity using invariance. There is no independence assertion about orbit samples.

Section 4: variable-input rows. Rows have support cells at a partition π_n; their adjoints have constant coefficients on a finer observation partition σ_n; the next support partition refines σ_n. The source stops and allocates *whole rows*. Private orthogonal coordinates make coefficients unique and stable under inheritance. A common-input Bessel estimate is converted, by maximal partial sums and Abel summation, to an estimate on depth-varying inputs. Smooth rows keep each carrier fixed across every cell at which it is used. Arbitrary adaptive carriers do not satisfy this premise.

Section 5: base construction. A successful fitting request gains actual projection energy. A finite path cap is chosen before its first-excess cuts; charged rows are estimated using the preceding variable-input theorem. Restarts have geometrically summable root lengths. The cyclic accuracy expansion uses two earliest increments, fixed actual input on each cell, and summable errors in accuracy, not in the original number of depths. High replacements require ordinary fixed-input Weyl, chart-tail, and Bessel estimates imported from [2]. Predictor lists precede high catalogs and opponents. Delayed projections are evaluated on the same F_s, not on a later F_{s+h}.

Section 6: localization, sides, and cores. Main catalogs and histories are distinguished from the larger mode catalog. Usable endpoints require whole historical pad cells and actual downstream use certificates. Certified variation couples all pad choices across times. The identity

$$L_t B_t' F_t - L_r B_r' F_r = (L_t - L_r) B_r' F_r + L_t (B_t' - B_r') F_r + L_t B_t' (F_t - F_r)$$

is used with a separate valid row estimate for each term. The frozen depth $u_b = b - 4K$, test depth $v_b = b - 2K$, and current block $[b, b + K)$ are different. Frozen mode rows are composed with a fixed lift and whole main-endpoint projections. Stored side vectors have full indices including group and block, and are estimated jointly. Near-band randomization retains the band sum inside the original cell. Sparse core counting is imported with persistent labels, same-cell side data, and the original tested-tuple constraints; marginal information alone cannot replace these hypotheses.

Section 7: completion. First-crossing cuts transfer only original summands. Parent realizations are frozen; children get newly constructed estimates, not proportional shares of arbitrary parent energies. Attempt roots have total length at most $4|R_0|/3$. A single $q > 2$ must satisfy both the source and companion parameter constraints. Summation over accuracy occurs before a finite-cell terminal residual is removed. This is different from asserting convergence of an infinite tree by inspection.

2.3 The companion proof remains a verification boundary

The companion [2] asserts, for fixed scalar compactly supported inputs and some $2 < q < 3$,

$$\sum_{I \in \mathcal{I}} |I|\, |H_I(\boldsymbol{f})| \le C_N(q) \int_{\mathbb{R}} \prod_{j=0}^{3} M_q f_j, \qquad \sup_N C_N(q) < \infty.$$

The headline $L^3 \times L^3 \times L^3 \to L^1$ assertion alone is not the package imported by M295. The relevant proof-level statements include Compression, Local detection, usable-endpoint Bessel separation, short-block comparison, sparse progression-test counting, and the order of absorption parameters.

The parts independently consulted here are specified in Section 12. Compression uses nested-prefix estimates for a *fixed common input*, followed by physical column control; it is not a license to substitute unrelated inputs into a Bessel inequality. The short-block estimate has original summation depth range N, Hilbert-stack L^2 sizes on every leaf, and finite sign-moment hypotheses. It costs

$$(1 + C_N(q))\, C_q (1 + \log(K+2))^C (K+2)^{11(q-2)/(q-1)}.$$

The exponent tends to zero as $q \downarrow 2$; a uniform theorem *at* $q = 2$ is not claimed.

M295's retained-cell estimate has the form

$$\begin{aligned} &C_q \eta^{\varepsilon} P^{C_4} |R| + (1 + C_N(q)) C_q P^{-1/2} G_0 (K+2)^{C_{19}(q-2)} |R| + O(e^{-k}|R|), \\ &\eta = e^{-k^{\theta}}, \quad P = \lceil \eta^{-\alpha} \rceil, \quad G_0 = \exp(C_q (1 + \log(2k))^C), \quad K \le C_q (1 + k + P)^{C_5}. \end{aligned} \tag{2.7}$$

The exponents must be fixed before q, then $C_{19} C_5 (q-2) < 1/8$, then $\alpha C_4 < \varepsilon/2$, then the starting accuracy. The corresponding companion constraint is imposed on the same q. Conditional on all stated uniform degree bounds and the companion absorption theorem, (2.7) closes algebraically. No numerical test verifies these uniform degree bounds or all the imported analytic constructions. No actionable contradiction was identified in the local deductions examined here; that is a bounded reading conclusion, not global proof certification.

3 Oscillation, arbitrary spaces, and all integers

3.1 A robust deterministic criterion

The next result is a useful quantitative refinement of the deterministic criterion in TR-item-168. It requires only integral maxima, not a full variation norm.

Theorem 3.1 (Oscillation with a linear defect)**.** *Let Z_k be finite measurable complex functions on a probability space. Suppose that for every deterministic w-block sequence $k_1 < \cdots < k_{w+1}$,*

$$\sum_{i=1}^{w} \mathbb{E} \max_{k_i \le \ell \le k_{i+1}} |Z_\ell - Z_{k_i}| \le b(w) < \infty.$$

For the measurable extended tail diameter $\Delta = \lim_{K\to\infty} \sup_{r,s\ge K} |Z_r - Z_s|$,

$$\mathbb{E}\Delta \le 2 \inf_{w\ge 1} \frac{b(w)}{w}. \tag{3.1}$$

In particular $b(w) = Cw^\alpha + \epsilon w$, $0 \le \alpha < 1$, gives $\mathbb{E}\Delta \le 2\epsilon$. At $\epsilon = 0$, Z_k converges almost everywhere.

Proof. For every fixed K, the triangle inequality gives

$$\sup_{\ell\ge K} |Z_\ell - Z_K| \ge \Delta/2.$$

Fix $0 < a < \mathbb{E}\Delta/2$, allowing any finite a if the expectation is infinite. Monotone convergence permits a deterministic $L > K$ with $\mathbb{E}\max_{K\le\ell\le L} |Z_\ell - Z_K| > a$. Repeat this choice w times, always using the last deterministic endpoint as the next start. Then $wa < b(w)$. This works for every w; let $a \uparrow \mathbb{E}\Delta/2$. If the expectation is infinite, the same argument contradicts finite $b(1)$. This proves (3.1). A finite-valued complex sequence with zero tail diameter is Cauchy and has a finite limit. □

The exceptional probability obeys $\mu\{\Delta > t\} \le 2\epsilon/t$. The premise never substitutes point-dependent crossing times into a deterministic-endpoint theorem. Constants growing like $o(w)$ are sufficient; a nonvanishing linear defect gives only approximate Cauchy behavior. Finite-horizon verification alone cannot determine Δ.

3.2 No hidden standardness assumption

For one fixed bounded triple, replace representatives on a common countable invariant null set if needed. The name map

$$\Xi(x) = \big(f_i(T^n x)\big)_{(i,n)\in\{1,2,3\}\times\mathbb{Z}}$$

takes values in a countable product of compact complex disks, a compact metrizable space. Its pushforward probability ν is invariant under the invertible coordinate shift S, and $\Xi \circ T = S \circ \Xi$. The coordinate functions reproduce every average in (1.1). Consequently a theorem for this standard Borel factor pulls back to the arbitrary original probability space. Strong mixing or weak mixing passes to this factor by pulling back measurable sets or bounded functions. This is a generated-factor argument, not an unjustified disintegration of the whole original space.

For O, even that reduction is unnecessary: finite sums, maxima, invariance, and integration suffice for the source transference. The reduction is useful when importing a norm theorem whose presentation uses a standard probability space.

3.3 Moment profiles and finite rational grids

For $c \in [1,2]$, let $h_c = c^{-1}\mathbf{1}_{(0,c]}$. Suppose that for each rational c and each integer $j \geq 1$ an admissible profile $\varphi_{c,j}$ satisfies $\|h_c - \varphi_{c,j}\|_1 \leq 1/j$. Section 6 constructs them explicitly. For $|F_n| \leq 1$, with $m_k = \lfloor c2^k \rfloor$,

$$\limsup_{k\to\infty} |A_{m_k} - V_k^{\varphi_{c,j}}| \leq \|h_c - \varphi_{c,j}\|_1 . \tag{3.2}$$

Indeed the normalization error is at most $1/(c2^k)$; the other error is bounded by $2^{-k}\sum_{n\in\mathbb{Z}} |h_c(n/2^k) - \varphi_{c,j}(n/2^k)|$, which tends to the L^1 norm by Riemann sums. No uniform profile constant in j is required.

Under O, Theorem 3.1 gives convergence of each smooth sequence. Intersect the countably many corresponding conull sets. Equation (3.2) makes each $A_{\lfloor c2^k \rfloor}$ Cauchy, with tail diameter at most $2/j$ for every j.

If A_N also has an L^2 limit L, each of these pointwise skeleton limits equals L: use an almost-everywhere convergent subsequence of its L^2 convergence. For each integer $s \geq 1$ use the finite grid $c_i = 1 + i/s$, $0 \leq i \leq s$. For $2^k \leq N < 2^{k+1}$, a lower grid index $n = \lfloor c_i 2^k \rfloor$ satisfies

$$0 \leq N - n < 2^k/s + 1, \qquad |A_N - A_n| \leq 2(N-n)/N \leq 2/s + 2^{1-k}.$$

The finitely many grid sequences converge to the same L. First let $k \to \infty$, then $s \to \infty$. Thus $A_N \to L$ along all integers.

Corollary 3.2 (A stronger conditional implication)**.** *If the universal premise* O *is correct, bounded triple averages converge almost everywhere on every invertible probability-preserving system, to their established L^2 limit.*

Proof. Use Host–Kra norm convergence [5, Theorem 1.1], the generated factor if necessary, and the preceding argument. □

This implication is materially stronger than the mixing headline. It makes verification of the claimed mixing-independent analytic scope particularly important. It is not an unconditional theorem proved by this report. In a general system L need not be a constant or the product of means. Merely knowing that different skeletons converge would not by itself identify their limits.

4 The conditional weak-mixing extension

4.1 Norm convergence without pointwise correlation decay

Weak mixing can be expressed by

$$\frac{1}{H}\sum_{h=1}^{H} |\left\langle U^{ah} f, f \right\rangle| \longrightarrow 0 \quad \text{if } \int f\, d\mu = 0, \quad a \in \mathbb{Z}\setminus\{0\}.$$

This standard correlation characterization follows from weak mixing of each nonzero power. It does not say that every individual correlation tends to zero.

Lemma 4.1 (Finite Hilbert-space inequality)**.** *For vectors $u_1, \ldots, u_N$ and an integer $H \geq 1$,*

$$\left\| \frac{1}{N}\sum_{n=1}^{N} u_n \right\|^2 \leq \frac{N+H-1}{N^2H^2}\Big[H\sum_{n=1}^{N}\|u_n\|^2 + 2\sum_{h=1}^{H-1}(H-h)\Re\sum_{n=1}^{N-h}\langle u_{n+h}, u_n\rangle \Big]. \tag{4.1}$$

Empty sums are zero.

Proof. Extend u_n by zero outside $[1, N]$. Sum the H-term moving windows: every u_n occurs exactly H times and at most $N + H - 1$ windows are nonzero. Cauchy–Schwarz bounds the squared norm of their sum by $N + H - 1$ times the sum of their squared norms. Expansion counts diagonal terms H times and lag-h terms $H - h$ times. Divide by N^2H^2. □

Proposition 4.2 (Weak-mixing product limit in norm)**.** *If T is weakly mixing, $a_1, \dots, a_d$ are distinct nonzero integers, and f_j are bounded, then*

$$\frac{1}{N}\sum_{n=1}^{N}\prod_{j=1}^{d} U^{a_j n} f_j \longrightarrow \prod_{j=1}^{d}\int f_j \, d\mu \quad \text{in } L^2.$$

Proof. Normalize the supremum norms and argue by induction on d. For $d = 1$, the mean ergodic theorem for the ergodic map T^{a_1} gives the conclusion. In the induction step first suppose one input is centered. Set $u_n = \prod_j U^{a_j n} f_j$ and $g_{j,h} = (U^{a_j h} f_j)\overline{f_j}$. For each fixed h, invariance and the induction hypothesis give

$$\begin{aligned}\frac{1}{N}\sum_{n=1}^{N-h}\langle u_{n+h}, u_n\rangle &= \int g_{1,h}\,\frac{1}{N}\sum_{n=1}^{N-h}\prod_{j=2}^{d} U^{(a_j-a_1)n} g_{j,h}\, d\mu \\ &\longrightarrow \prod_{j=1}^{d}\int g_{j,h}\, d\mu =: c_h.\end{aligned}$$

The remaining slopes are distinct and nonzero. The finitely missing terms have norm $O(h/N)$, with h fixed. If f_{j_0} is centered, $|c_h| \le |\left\langle U^{a_{j_0} h} f_{j_0}, f_{j_0}\right\rangle|$, so $H^{-1}\sum_{h\le H}|c_h| \to 0$. Take $N \to \infty$ at fixed H in (4.1):

$$\limsup_N \left\| N^{-1}\sum u_n \right\|_2^2 \le H^{-1} + \frac{2}{H}\sum_{h=1}^{H-1}|c_h|.$$

Now let $H \to \infty$. Expand arbitrary inputs into their means and centered parts. Every nonconstant term has a centered factor and tends to zero; the constant term is the claimed product. □

This is the classical weak-mixing norm mechanism, supplied explicitly to pin down the quantifiers. There is no limit with $h = h(N)$, no uniform correlation rate, and no stationarity assumption for u_n.

Corollary 4.3 (Conditional pointwise product limit)**.** *Assuming* O*, (1.3) holds for every invertible weakly mixing probability-preserving transformation and every bounded triple, along all N.*

Proof. Use Proposition 4.2 for $a_1, a_2, a_3 = 1, 2, 3$ and the profile and grid argument of the preceding section. □

The source proves its norm identification under strong mixing and mentions the established weak-mixing norm theory in its introduction. The corollary isolates a consequence of its stronger analytic scope; it does not remove that scope's proof obligation.

5 Integrability through the endpoint

The following extension does not enlarge the difficult analytic estimate to unbounded inputs. It uses the bounded theorem and elementary positive maximal bounds.

Theorem 5.1 (Conditional L^p extension)**.** *Suppose that on a fixed probability-preserving system the bounded triples in* (1.1) *converge almost everywhere to their product of means. Let* $1 \le p_i \le \infty$*, and put*

$$0 < s = \sum_{i=1}^{3} p_i^{-1} \le 1, \qquad r = s^{-1}.$$

Then for $f_i \in L^{p_i}$*, the same averages converge to the product of means almost everywhere and in* L^r*. Thus, under* O*, this applies to every weakly mixing system, including every mixing system. When* $s = 0$*, the bounded conclusion applies; no* L^∞ *norm convergence is asserted.*

Proof. Write $M_i g = \sup_{N\ge 1} N^{-1} \sum_{n=1}^{N} g(T^{in}x)$ for nonnegative g. The positive maximal ergodic inequality gives $\mu\{M_i g > t\} \le \|g\|_1 / t$, without ergodicity. Let $I = \{i : p_i < \infty\}$, $m = |I|$, and $K = \prod_i \|f_i\|_{p_i}$. If $K = 0$, countable invariance makes all products zero almost everywhere. Otherwise finite-average Hölder, with a constant factor when the reciprocal sum is below one, yields

$$A^*(\boldsymbol{f}) := \sup_N |A_N(\boldsymbol{f})| \le \prod_{i\notin I} \|f_i\|_\infty \prod_{i\in I} [M_i(|f_i|^{p_i})]^{1/p_i}.$$

If $M_i(|f_i|^{p_i}) \le t \|f_i\|_{p_i}^{p_i}$ for every finite index, then $A^* \le K t^s$. Taking $t = (\lambda/K)^{1/s}$ and a union bound proves

$$\mu\{A^*(\boldsymbol{f}) > \lambda\} \le \min\{1, m(K/\lambda)^r\}. \tag{5.1}$$

Truncate each finite-p_i input radially at height J, leaving the L^∞ inputs unchanged; call the resulting triple $\boldsymbol{f}^{(J)}$. Its coordinate errors tend to zero in their respective norms and its norms are bounded by the original ones. Telescoping the product expresses $A_N(\boldsymbol{f}) - A_N(\boldsymbol{f}^{(J)})$ as at most three multilinear averages, each with one norm-small error and bounded remaining norms. Equation (5.1), with threshold divided by three, makes the probability of the maximal difference exceeding any fixed positive threshold tend to zero.

Let $\Delta(\boldsymbol{f})$ be the tail diameter in N. The approximant converges on a conull set, and $\Delta(\boldsymbol{f}) \le 2\sup_N |A_N(\boldsymbol{f}) - A_N(\boldsymbol{f}^{(J)})|$ there. Thus $\Delta(\boldsymbol{f}) = 0$ almost everywhere by taking $J \to \infty$ at each rational threshold. Countably many orbit evaluations of finite functions are finite almost everywhere, so the pointwise limit exists.

For its norm and identification, Hölder in space and Minkowski give, for every N,

$$\|A_N(\boldsymbol{f})\|_r \le \prod_i \|f_i\|_{p_i} = K. \tag{5.2}$$

The same telescoping argument now gives a uniform-in-N L^r approximation error tending to zero. For bounded approximants, almost-everywhere convergence and bounded convergence imply L^r convergence. Their product means tend to $p(\boldsymbol{f})$, since L^{p_i} convergence implies L^1 convergence on a probability space. The triangle inequality proves $A_N(\boldsymbol{f}) \to p(\boldsymbol{f})$ in L^r, including $r = 1$. A subsequence identifies the already existing pointwise limit. □

Proposition 5.2 (Strong maximal control below the reciprocal endpoint)**.** *If* $0 < s < 1$*,* $r = 1/s > 1$*, then unconditionally on every probability-preserving system*

$$\|A^*(\boldsymbol{f})\|_r \le \frac{r}{r-1} \prod_i \|f_i\|_{p_i}. \tag{5.3}$$

Proof. For the finite indices set $q_i = sp_i$. Each $q_i \geq 1$ and $\sum_{i\in I} 1/q_i = 1$. Finite-average Hölder bounds A^* by the product of $[M_i(|f_i|^{q_i})]^{1/q_i}$ and the L^∞ factors. Spatial Hölder and the scalar strong L^r maximal bound, whose constant is $r/(r-1)$, give

$$\|A^*\|_r \leq \prod_{i\in I} \|M_i(|f_i|^{q_i})\|_r^{1/q_i} \prod_{i\notin I} \|f_i\|_\infty \leq \left(\frac{r}{r-1}\right)^{\sum 1/q_i} K.$$

For completeness, use the scalar maximal operator M^0 with averages starting at index zero. Integrating its refined maximal ergodic inequality $t\,\mu\{M^0 g > t\} \leq \int_{\{M^0 g>t\}} g\,d\mu$ gives $\|M^0 g\|_r^r \leq [r/(r-1)] \int g(M^0 g)^{r-1}$, followed by Hölder and finite truncation before passage to the limit. The operator M_i, whose averages start at one, is M^0 for the map T^i applied to $g \circ T^i$; invariance preserves its input norm. □

The constant diverges as $r \downarrow 1$; (5.1), not (5.3), is used at $s = 1$. No claim is made for $s > 1$, for a new trilinear variation estimate, or for a quantitative mixing rate. A maximal estimate by itself does not prove convergence: the dense bounded convergence class remains essential.

Remark 5.3 (An optional optimized weak constant)**.** Put $\theta_i = (1/p_i)/s$ on the finite indices. Instead of equal thresholds, use $M_i(|f_i|^{p_i}) \leq t_i \|f_i\|_{p_i}^{p_i}$. For $L = \lambda/K$, minimize $\sum t_i^{-1}$ subject to $\prod t_i^{1/p_i} = L$. The minimizer has $t_i^{-1} = c\theta_i$, and $c = L^{-r} \prod \theta_i^{-\theta_i}$. Thus m in (5.1) can be replaced by $\exp(-\sum \theta_i \log \theta_i) \leq m$. This is a threshold optimization, not independence of the maximal functions.

6 An explicit and sharp moment bank

6.1 Four translated bumps

Fix an even nonnegative C^∞ Gevrey bump ρ supported in $[-1, 1]$, with $\int \rho = 1$, and write $\mu_2 = \int x^2 \rho(x)\,dx$. Let ψ be any fixed smooth compactly supported mass-one function with moments $m_j = \int x^j \psi(x)\,dx$, $1 \leq j \leq 3$. For $R > 4$, define

$$\begin{aligned} A &= -m_1/R, & B &= (-m_3 + 3\mu_2 m_1)/R^3, \\ a &= -2m_2/(3R^2), & b &= (4B - A)/6, & c &= 4(A - B)/3. \end{aligned} \tag{6.1}$$

With $\rho_y(x) = \rho(x - y)$, set

$$q_R = (a + b)\rho_R + (a - b)\rho_{-R} + (-a + c)\rho_{R/2} + (-a - c)\rho_{-R/2}. \tag{6.2}$$

Proposition 6.1 (Exact correction and total-variation budget)**.** *The function $\varphi_R = \psi + q_R$ has mass one and moments one through three equal to zero. Moreover*

$$\|q_R\|_1 \leq \frac{3|m_1|}{R} + \frac{8|m_2|}{3R^2} + \frac{4|m_3 - 3\mu_2 m_1|}{R^3}. \tag{6.3}$$

For fixed moments, the leading constant $3|m_1|$ is asymptotically optimal among signed corrections supported in $[-R - 1, R + 1]$ with these mass and moment requirements.

Proof. The four coefficients sum to zero. A translated even bump has first three moments $y, y^2 + \mu_2, y^3 + 3\mu_2 y$. The correction moments therefore are

$$R(2b + c) = -m_1, \qquad \frac{3}{2} a R^2 = -m_2, \qquad R^3(2b + c/4) - 3\mu_2 m_1 = -m_3,$$

using $2b + c = A$ and $2b + c/4 = B$. For $R > 4$ the four supports are disjoint, so

$$\|q_R\|_1 = |a + b| + |a - b| + | - a + c| + | - a - c| \le 4|a| + 2|b| + 2|c| \le 4|a| + 3|A| + 4|B|.$$

This is (6.3).

For any signed measure q supported in $[-R - 1, R + 1]$ with the required moments, test against $p_R(x) = 3x/(R + 1) - 4(x/(R + 1))^3$. The identity

$$(3z - 4z^3)^2 - 1 = (z^2 - 1)(4z^2 - 1)^2$$

shows $|p_R| \le 1$ on this interval, without any appeal to an unproved extremal-polynomial assertion. Thus

$$\|q\|_{\mathrm{TV}} \ge \left| -\frac{3m_1}{R+1} + \frac{4m_3}{(R+1)^3} \right|. \tag{6.4}$$

Multiply the lower bound and (6.3) by R and let $R \to \infty$. Both give $3|m_1|$. The conclusion is substantive when $m_1 \neq 0$; it makes no claim about the optimal next term when $m_1 = 0$. □

6.2 Application to the source profiles

Take $\psi = h_c * \rho_\delta$, where $\rho_\delta(x) = \delta^{-1}\rho(x/\delta)$. Translation continuity of an interval indicator gives

$$\|\psi - h_c\|_1 \le \frac{2\delta}{c} \int |y|\rho(y)\, dy.$$

Its moments are finite; specifically $m_1 = c/2$, $m_2 = c^2/3 + \delta^2\mu_2$, and $m_3 = c^3/4 + (3c/2)\delta^2\mu_2$. Choose δ first, then R so that (6.3) is as small as required. This produces $\|\varphi_R - h_c\|_1 \to 0$ with exact moments. Differentiation commutes with convolution, and all four corrections are translates of the same Gevrey bump, so each individual φ_R satisfies the derivative condition (2.1) with a finite constant.

This replaces the source's abstract four-by-four Vandermonde correction by an explicit bank and error budget. It does not remove support dependence: the support radius grows like R, and the local-kernel and oscillation constants may grow with it. The all-integer proof needs only countably many finite constants, not a common bound as the profile error tends to zero.

The escaping support is unavoidable. If both h_c and φ are supported in $[-S, S]$ and φ has first moment zero,

$$\|\varphi - h_c\|_1 \ge c/(2S).$$

Also a nonnegative smooth mass-one function cannot have second moment zero: $x^2\varphi \ge 0$ would force its support into a single point. Signed profiles are a mathematical necessity here, not a normalization convenience.

7 Local row refinements with unchanged hypotheses

7.1 The physical catalog and chronology

Let R be a finite dyadic root and $\mathcal{H} = \mathbb{C} \oplus \ell^2(\mathcal{V})$. A label v is born on a dyadic cell J_v and has column $\Phi_v = (\phi_v, \gamma_v e_v)$, restricted unchanged to descendants. Assume

$$\|\Phi_v(y)\|_{\mathcal{H}} \le D, \qquad |\gamma_v| \ge D^{-1}, \qquad \sum_v \mathbf{1}_{J_v}(y) \le D, \qquad D \ge 2.$$

All available masks are inherited, including labels not currently used. For constant coefficient combinations, the private coordinates and Cauchy–Schwarz give

$$D^{-1}\|a\|_{\ell^2} \le \left\|\sum_v a_v \Phi_v(y)\right\|_{\mathcal{H}} \le D^{3/2}\|a\|_{\ell^2}. \tag{7.1}$$

The upper bound uses at most D available labels; the lower bound uses distinct private coordinates.

In round n, rows $A_\nu : L^2(R;\mathcal{H}) \to \mathcal{Y}_\nu$ are supported on whole cells of π_n. On every cell of σ_n, each $A_\nu^* c$ is a constant-coefficient combination of available columns. Assume σ_n refines π_n, and π_{n+1} refines σ_n. Crucially,

$$\sum_\nu \|A_\nu z\|^2 \le B^2\|z\|_2^2 \quad \text{for every one common } z \in L^2(R;\mathcal{H}). \tag{7.2}$$

The realized rows may depend on the schedules, but are held fixed while (7.2) is tested on arbitrary z. This is stronger than observing the estimate on the particular schedule used by the construction.

7.2 The maximal-to-depth estimate

For completeness we record the local mechanism used for the refinements. It is the source's Section 4 argument, not an invocation of its global Hilbert-transform theorem. Set

$$L_D = 1 + \lceil 5\log_2 D\rceil, \qquad C_D = \frac{8}{3}\sqrt{9 + L_D^2} \le \frac{40}{3}\log_2(2D).$$

For arbitrary row tests c_ν, put $E_c = \sum_\nu \|c_\nu\|^2$, $g_n = \sum_{\nu \text{ in round } n} A_\nu^* c_\nu$, and $M = \max_j \left\|\sum_{n\le j} g_n\right\|_{\mathcal{H}}$, with any fixed consecutive range of rounds.

Lemma 7.1 (Local maximal estimate)**.** *Under the preceding catalog, support, observation, and Bessel hypotheses,*

$$|\{M > \lambda\}| \le \min\{|R|, C_D^2 B^2 E_c/(4\lambda^2)\}, \qquad \|M\|_1 \le C_D B |R|^{1/2} E_c^{1/2}.$$

Proof. Here are the stopping details that make the depth independence legitimate. Put $T = 2^{\lceil 5\log_2 D\rceil}$ and $\tau = \lambda/(4D^{3/2})$. Accumulate adjoint rows in a current bucket on each path. Assign a whole support cell's rows to its current bucket before observing their updated coefficients. A hit occurs when that bucket's coefficient vector has norm greater than τ; advance the bucket and stop after T hits. States before assignment are constant on π_n; updates are constant on σ_n; the next support partition refines this. Thus each original row is assigned wholly to one bucket or omitted, never spatially fragmented after observation.

Write v_i for the sum of adjoint tests in bucket i. The bucket row sets are disjoint. Duality of (7.2), applied separately to every bucket, gives $\sum_i \|v_i\|_2^2 \le B^2 E_c$. On a path that reaches T hits, each completed bucket has physical norm at least $D^{-1}\tau$, by (7.1); its value cannot change later on that path. Hence the measure of such paths is at most $16D^5 B^2 E_c/(T\lambda^2) \le 16B^2E_c/\lambda^2$.

On every other path, a partial round sum consists of a prefix of completed buckets plus an unfinished bucket of norm at most $D^{3/2}\tau = \lambda/4$. Let M_T be the maximum over prefixes of the full bucket sequence. Decompose a prefix into dyadic index blocks, at most one at each of the L_D lengths. At one length, the row sets of these blocks are disjoint, so the sum of their squared L^2 norms is at most B^2E_c. Cauchy–Schwarz over lengths yields $\|M_T\|_2 \le L_D B E_c^{1/2}$. Consequently

$$|\{M > \lambda\}| \le \frac{16B^2E_c}{\lambda^2} + \frac{16L_D^2B^2E_c}{9\lambda^2} = \frac{C_D^2B^2E_c}{4\lambda^2}.$$

Integrating the minimum of this bound and $|R|$ over $\lambda > 0$ gives the L^1 assertion. The same construction works on a subrange of rounds. □

Proposition 7.2 (Unequal block budgets)**.** *Let F_s be a finite depth schedule with deterministic consecutive blocks S_b, and assume pointwise*

$$\sup_{s\in\mathsf{S}_b}\|F_s\| + \sum_{s,s+1\in\mathsf{S}_b}\|F_{s+1}-F_s\| \le a_b.$$

Let the nondecreasing read indices s_n be common to every row in round n. Then

$$\left(\sum_n \sum_{\nu \text{ in round } n} \|A_\nu F_{s_n}\|^2\right)^{1/2} \le C_D B|R|^{1/2}\left(\sum_b a_b^2\right)^{1/2}. \tag{7.3}$$

Proof. Pair the output with arbitrary c_ν. Reads in one block occupy a consecutive range of rounds, say $p,\dots,q$. Pointwise Abel summation gives, with $S_n = \sum_{r=p}^{n} g_r$,

$$\sum_{n=p}^{q}\langle F_{s_n}, g_n\rangle = \langle F_{s_q}, S_q\rangle + \sum_{n=p}^{q-1}\langle F_{s_n}-F_{s_{n+1}}, S_n\rangle.$$

Its absolute value is at most $a_b \max\|S_n\|$, since nondecreasing reads traverse each schedule edge at most once. Lemma 7.1 bounds its integral by $a_b C_D B\sqrt{|R|}\sqrt{E_b}$. Cauchy–Schwarz over blocks and Hilbert direct-sum duality prove (7.3). □

The source normalization $a_b = W^{-1/2}$ gives $\sum_b a_b^2 = 1$. Unequal error or regularity allocations are allowed without introducing the number of blocks as a separate loss. Random or spatially varying block partitions are not covered by this statement.

7.3 A square-root delay bound

Let P_h be direct sums of orthogonal projections onto inherited constant-coefficient catalogs on depth-h cells. They are nested. Assume the source block normalization, and write $\|z\|_{2,I}^2 = |I|^{-1}\int_I \|z\|^2$. For any selected depth-s cells $\mathcal{U}_s$, the source estimates the delayed expression by $LC_D\sqrt{|R|}$. There is a sharper consequence of the same hypotheses.

Proposition 7.3 (Square-root delay)**.** *For every integer $L \ge 1$, retaining only indices for which both endpoints exist,*

$$\left(\sum_s \sum_{I\in\mathcal{U}_s} |I|\ \|(P_{s+L}-P_s)F_s\|_{2,I}^2\right)^{1/2} \le \sqrt{L}\, C_D\sqrt{|R|}. \tag{7.4}$$

For $L = 0$ the expression is zero.

Proof. If intervals $[r_j, t_j]$ have disjoint interiors in increasing order, $P_{t_j} - P_{r_j}$ are orthogonal projections with pairwise orthogonal ranges. Split each row over its depth-r_j support cells. The split rows retain the common-input Bessel constant one: their outputs on disjoint support cells are orthogonal. Their adjoints are constant-column combinations on depth-t_j observation cells, by inheritance. Proposition 7.2 therefore gives

$$\sum_j \left\|(P_{t_j}-P_{r_j})F_{s_j}\right\|_2^2 \le C_D^2|R|$$

whenever the reads s_j are nondecreasing. Divide the intervals $[s, s+L]$ into the L classes of s modulo L. Each is such a disjoint ordered family, with increasing reads s. Sum the L squared bounds. Restriction to the disjoint selected cells at each s can only decrease the energy. □

This improves the source's use of Minkowski over L increments. It also extends to intervals with edge overlap m and increasing reads: greedy coloring gives the factor $\sqrt{m}$. A square-root multiplicity cost is natural. For projections jumping from zero to a one-dimensional range at depth h, a fixed unit vector in that range contributes energy one for each of the L indices $h - L \le s < h$; total energy is L. The refinement does not make delayed states small uniformly in unbounded L, and approximate locality does not create exact projection inheritance.

7.4 A logarithmic-3/2 smooth-row bound

Retain every hypothesis of the source's Lemma `lem:smooth-depth`: finitely many fixed carriers m_ν, $|m_\nu| \le 1$, at most one row per (I, ν), mean-zero C^1 bumps $\omega_{I,\nu}$ supported on I, with

$$\|\omega_{I,\nu}\|_\infty + |I| \left\|\omega'_{I,\nu}\right\|_\infty \le A,$$

and the physical common-input bound

$$\sum_{I,\nu} \left| |I|^{-1/2} \int_I m_\nu \omega_{I,\nu} f \right|^2 \le B_H^2 \, \|f\|_2^2 \, .$$

Set $d = 2 + \#\{\nu\}$ and $h_0 = \lceil 10 \log_2(2d) \rceil$. The input is f_s with the source normalization, embedded as $F_s = (f_s, 0)$.

Proposition 7.4 (Disjoint residue outputs)**.** *The depth-varying row norm is at most*

$$\begin{aligned} &C_d \left[\sqrt{h_0 + 1}(B_H + 3A) + 2A\sqrt{d}\, 2^{-h_0} \sqrt{h_0 + 3} \right] \sqrt{|R|} \\ &\qquad \le 227 (B_H + A) \, [\log_2(2d)]^{3/2} \sqrt{|R|}. \end{aligned} \tag{7.5}$$

The source's displayed bound has a logarithmic-square loss.

Proof. Let E_j be dyadic conditional expectation. The normalized bump at a depth-s cell has martingale pieces

$$q_{I,\nu,h} = |I|^{-1/2} (E_{s+h} - E_{s+h-1}) \overline{\omega_{I,\nu}}, \qquad h \ge 1.$$

They satisfy $\|q_{I,\nu,h}\|_2 \le 2A 2^{-h}$. For a fixed ν, h, different cells have either disjoint support or orthogonal martingale depths; multiplication of the common test input by $\overline{m}_\nu$ is a contraction. Cauchy–Schwarz over the carriers gives the physical Bessel norm $2A\sqrt{\#\{\nu\}}\, 2^{-h}$. Augment with columns $(\overline{m}_\nu, e_\nu)$. The private-coordinate operator has norm at most $2A 2^{-h}$. Combining the orthogonal physical and private input summands by Cauchy–Schwarz gives augmented tail norm at most $2A\sqrt{\#\{\nu\} + 1}\, 2^{-h} \le 2A\sqrt{d}\, 2^{-h}$. For the main part $\sum_{h \le h_0} q_{I,\nu,h}$, subtracting the small tail from the original physical row bank and adding the private coordinates gives augmented norm at most $B_H + 3A$; the private components cost at most $2A$. These are the same ordinary, fixed-input estimates used by the source.

At a fixed lag h, separate support depths modulo $h + 1$. Within a residue class the observation depth $s + h$ precedes the next support depth, so the row chronology is valid and Proposition 7.2 applies with catalog budget d. The main part uses $h_0 + 1$ classes. Different residue classes occupy *disjoint coordinates of the output row Hilbert space.* Their squared norms therefore add: their costs are $\sqrt{h + 1}$ and $\sqrt{h_0 + 1}$, not $h + 1$ and $h_0 + 1$. Different lags contribute to the same original row, so at that stage we do use the triangle inequality. The result is

$$C_d \left[\sqrt{h_0 + 1}(B_H + 3A) + 2A\sqrt{d} \sum_{h > h_0} \sqrt{h + 1}\, 2^{-h} \right] \sqrt{|R|}.$$

The weights $2^{-h}/2^{-h_0}$, $h > h_0$, sum to one and have mean $h + 1 = h_0 + 3$. Concavity of the square root yields the first line of (7.5).

Put $\ell = \log_2(2d) \geq 2$. Then $h_0 + 1 \leq 11\ell$, $h_0 + 3 \leq 12\ell$, and $\sqrt{d}\, 2^{-h_0} \leq 1$. Since $3\sqrt{11} + 2\sqrt{12} < 17$ and $C_d \leq (40/3)\ell$, the coefficient is below $(680/3)(B_H + A)\ell^{3/2} < 227(B_H + A)\ell^{3/2}$. □

This is a concrete improvement of a local estimate with precisely the same carrier and row conditions. It neither establishes the ordinary Bessel hypothesis for new carriers nor verifies the companion's mode-separation or core-counting theorems. In the source hierarchy, fixed logarithmic losses already fit polynomial budgets; a better local exponent does not by itself close a previously unproved global step.

8 Exact transference and a second-order error

8.1 All four cancellations are retained

Let φ satisfy (2.1), $\varphi_2(t) = \frac{1}{2}\varphi(t/2)$, and $b = \varphi_2 - \varphi$. Its moments of orders $0, 1, 2, 3$ vanish. Consequently its fourth primitive g, taken from $-\infty$, is compactly supported. Choose a compactly supported smooth u of integral one, a sufficiently large fixed dyadic D, and set

$$W(z, \sigma) = \prod_{j=0}^{3} (\partial_\sigma - j\partial_z)[u(z) D^{-3} g(D\sigma)].$$

Commutativity of the derivatives gives the exact identity

$$\begin{aligned} W = {} & Du(z) g^{(4)}(D\sigma) - 6u'(z) g^{(3)}(D\sigma) + 11 D^{-1} u''(z) g''(D\sigma) \\ & - 6D^{-2} u'''(z) g'(D\sigma). \end{aligned} \tag{8.1}$$

For the j-th line $z = v - j\sigma$, the operator $\partial_\sigma - j\partial_z$ is its total derivative. Compact support makes that line integral zero. Integrating in z instead kills the derivatives of u, leaving $\int W(z, \sigma)\, dz = Db(D\sigma)$. After spatial rescaling $r = Da$, this is the desired scale difference $a^{-1} b(t/a)$. The support is placed strictly inside all four coordinate cells by choosing D large relative to the fixed profile support. Thus line cancellation is exact, not merely small in a finite moment table.

8.2 Symmetry improves lattice planting

The source places integer data on intervals $[m - e, m + e]$ with $e = 1/100$. A nonzero real progression through four such intervals has integer centers $m, m + n, m + 2n, m + 3n$: any discrepancy is an integer of absolute value less than one. With $x = m + \xi$, $t = n + \eta$, the offset region is

$$Q_e = \{(\xi, \eta) : |\xi + j\eta| \leq e,\ 0 \leq j \leq 3\}.$$

Its middle constraints follow from convexity of the two endpoints. The change of variables $u = \xi$, $v = \xi + 3\eta$ maps it onto $[-e, e]^2$, with Jacobian $1/3$. Writing $a_e = |Q_e| = 4e^2/3$, we obtain

$$\frac{1}{a_e} \int_{Q_e} \eta = 0, \qquad \frac{1}{a_e} \int_{Q_e} \eta^2 = \frac{2e^2}{27}, \qquad |\eta| \leq 2e/3. \tag{8.2}$$

Proposition 8.1 (Quadratic scale error)**.** *For $\Delta_k(t) = 2^{-k} b(t/2^k)$, any integer n, and $k \geq 0$,*

$$\left| \frac{1}{a_e} \int_{Q_e} \Delta_k(n + \eta)\, d\xi\, d\eta - \Delta_k(n) \right| \leq \frac{e^2}{27} \left\| b'' \right\|_\infty 2^{-3k}. \tag{8.3}$$

If $\operatorname{supp} b \subset [-B, B]$*, integer spatial support has length* L*, and the absolute value of each coefficient product is at most* M_0*, then the planted continuous sum* I_k *and its discrete counterpart* D_k *satisfy*

$$|I_k - a_e D_k| \le a_e \frac{e^2}{27} \left\|b''\right\|_\infty (2B + 4e/3 + 1)\, L M_0\, 2^{-2k}. \tag{8.4}$$

Proof. Taylor's formula has remainder at most $\frac{1}{2}\left\|\Delta_k''\right\|_\infty \eta^2$. The linear term integrates to zero by (8.2); $\left\|\Delta_k''\right\|_\infty = 2^{-3k}\left\|b''\right\|_\infty$. This proves (8.3). Only $|n| \le B2^k + 2e/3$ can contribute to the difference; the number of such integers is at most $(2B + 4e/3 + 1)2^k$. Sum over them and the L spatial centers. □

The geometric tail is $\sum_{k \ge k_-} 2^{-2k} = (4/3)2^{-2k_-}$. The source uses a valid first-derivative 2^{-k} summed error; symmetry yields the stronger second-order error above, with a second-derivative constant instead. This is a local discretization improvement, not an oscillation theorem.

8.3 Linearization and the final order of limits

For a fixed finite endpoint list and each integer spatial point, choose the first maximizing index in each block and align its complex phase. The source uses a prefix schedule $\alpha_i/(4\sqrt{w})$; $\alpha_i/(2\sqrt{w})$ also satisfies (2.5), because its supremum and its single internal jump each cost at most $1/(2\sqrt{w})$. This halves the final linearization factor with the rest of the local estimate unchanged. The schedules are deterministic functions on the integer model after they have been selected; their number of depth blocks is w.

For orbit transference, fix these endpoints first and pad every sampled orbit far enough to contain all shifts at the largest scale. Apply the integer inequality, integrate in x, and use invariance to make each spatial translate have the same integral. Divide by the spatial length and let it tend to infinity. The boundary-to-length ratio then vanishes for these fixed endpoints. A bound uniform in the endpoints can subsequently be stated because the analytic constant never depended on them. Letting the largest endpoint grow while keeping an unpadded finite orbit length fixed would not justify this passage.

9 Physical Ram certificates and finite-model limits

9.1 Gram coordinates are not coefficient energy

TR248 [4, Sections 2–4; moment-assembly appendix] makes the relevant finite-dimensional distinction explicit. Let $\Phi_1, \dots, \Phi_d$ be linearly independent columns on a declared probability patch, and $G_{vw} = \int \langle \Phi_w, \Phi_v \rangle \, d\mu$, in the corresponding matrix convention. The squared physical norm of a synthesized field is $a^* G a$, not $a^* a$. Private coordinates contribute a positive diagonal $\operatorname{diag}(|\gamma_v|^2)$ to G, in addition to the scalar Gram. For row-coordinate matrix B, the exact model Bessel certificate is

$$B^* B \preceq C^2 G. \tag{9.1}$$

Equivalently $\left\|BG^{-1/2}\right\| \le C$. With positive sampled weights Q, physical coordinates are $Q^{1/2}\Phi a$; unweighted QR would certify a different inner product.

For a concrete exact example on normalized Lebesgue measure on $[-1, 1]$, take columns $\Phi_1 = (1, \frac{1}{4}e_1)$, $\Phi_2 = (x, \frac{1}{4}e_2)$, and

$$G = \begin{pmatrix} 17/16 & 0 \\ 0 & 19/48 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1/2 \\ 0 & 1/3 \end{pmatrix}.$$

Then

$$2G - B^*B = \begin{pmatrix} 9/8 & -1/2 \\ -1/2 & 31/72 \end{pmatrix}, \qquad \det(2G - B^*B) = 15/64 > 0.$$

The positive first pivot and determinant give the exact bound (9.1) with $C = \sqrt{2}$. This is a certificate for the stated model, not a test of the source's enormous adaptive catalog.

If the actual row Riesz representatives all lie in the model space V, then the rows annihilate $V^\perp$, and (9.1) gives their full Hilbert-space bound. Without that premise one must also bound $\|A(I - \Pi_V)\|$; knowing only the matrix on V leaves this operator entirely uncontrolled. For example, $\|A\Pi_V\| \le C_0$ and $\|A(I - \Pi_V)\| \le \epsilon$ imply $\|A\| \le C_0 + \epsilon$ by the triangle inequality. A finite list of tests alone cannot prove the second premise.

On exponential-polynomial Ram atoms,

$$\phi_j(t) = e^{st}t^j, \qquad G_{jk} = M_{j+k}(2\Re s), \qquad M_\ell(z) = \frac{d^\ell}{dz^\ell}\int e^{zt}\, d\mu(t),$$

so the generating function supplies actual Gram entries. For Lebesgue measure on $[-1, 1]$, $M_0(z) = 2\sinh(z)/z$, with continuous value 2 at zero. Expanding the exponential through degree L, uniformly for $|z| \le R_0$, bounds each moment remainder by $2e^{R_0}R_0^{L+1}/(L+1)!$. This follows directly from Taylor's remainder and $|t| \le 1$. It permits rigorous entry enclosures and rational LDL^* certificates. It is not a theorem that all nonlinear chart atoms or arbitrary smooth cutoffs have geometrically small tails.

9.2 Leakage as an actual operator

Suppose windows w_p obey $\sum_p w_p^2 = 1$ and Π_p are local orthogonal projections. With M_p multiplication by w_p, TR248's retained analysis map satisfies

$$F^*F = I - \mathcal{L}, \qquad \mathcal{L} = \sum_p M_p(I - \Pi_p)M_p, \qquad \langle \mathcal{L}f, f\rangle = \sum_p \|(I - \Pi_p)(w_p f)\|_2^2 . \tag{9.2}$$

Every identity is on the specified physical space. For an isometric model embedding $U_V : \mathbb{C}^d \to V$,

$$(FU_V)^*(FU_V) = I - U_V^*\mathcal{L}U_V .$$

Thus a certified bound $\|U_V^*\mathcal{L}U_V\| \le \delta^2 < 1$ gives model singular values at least $\sqrt{1 - \delta^2}$. This controls reconstruction on V; it does not eliminate an unknown $V^\perp$ component.

A closely related screening certificate uses two nested projections $P \le Q$:

$$\|(Q - P)f\|_2^2 = \|Qf\|_2^2 - \|Pf\|_2^2 . \tag{9.3}$$

If the measured energy difference is at most η, then for any bounded test b,

$$|\,\langle (Q - P)f, b\rangle\,| \le \sqrt{\eta}\ \|b\|_2 \le \sqrt{\eta}\ \|b\|_\infty .$$

This follows because $Q - P$ is an orthogonal projection. An approximate Gram model requires an additional enclosure for the energy difference. Small residual energy does not imply that a label is exactly absent or that an active interaction is exactly zero. Such a conclusion requires a stated positive detection gap.

9.3 Nonlinear products and finite orbit errors

Let all f_i be one-bounded, and suppose bounded physical approximants $g_{i,n}$ satisfy $\|U^{in} f_i - g_{i,n}\|_2 \le \epsilon_{i,n}$. A decoded field may first be projected pointwise onto the complex unit disk: this cannot increase its distance to the true one-bounded field. Telescoping three physical factors gives

$$\left\| A_N(\boldsymbol{f}) - N^{-1} \sum_{n=1}^{N} \prod_{i=1}^{3} g_{i,n} \right\|_2 \le \frac{1}{N} \sum_{n=1}^{N} \sum_{i=1}^{3} \epsilon_{i,n}. \tag{9.4}$$

This is the direct task-specific use of a Ram reconstruction bound. Multiplying coefficient entries is generally not physical multiplication; the reconstruction and product map must be declared.

If U, V are Koopman unitaries on the same probability space and $\|U - V\|_{2\to 2} \le \epsilon$, telescoping powers gives $\left\|(U^k - V^k) f\right\|_2 \le k\epsilon$ for one-bounded f. Equation (9.4) then gives

$$\left\| A_N^U(\boldsymbol{f}) - A_N^V(\boldsymbol{f}) \right\|_2 \le 3(N+1)\epsilon.$$

This is TR-item-168's finite-horizon budget, with its strong operator-norm premise retained. The error grows with N; it cannot be called an all-scale convergence certificate. The individual orbit-error premise in (9.4) is often more usable, but still needs verification.

9.4 A bounded-input L^1 approximation refinement

For two one-bounded triples $\boldsymbol{f}, \boldsymbol{g}$,

$$\sup_N |A_N(\boldsymbol{f}) - A_N(\boldsymbol{g})| \le \sum_{i=1}^{3} M_i |f_i - g_i|.$$

Set $e_i = \|f_i - g_i\|_1$. A union bound with thresholds $\tau_i > 0$, $\sum \tau_i = t$, and the positive weak-$(1,1)$ inequality gives

$$\mu\{\sup_N |A_N(\boldsymbol{f}) - A_N(\boldsymbol{g})| > t\} \le \sum_i e_i/\tau_i.$$

Choose $\tau_i = t\sqrt{e_i}/\sum_j \sqrt{e_j}$, omitting zero terms, to obtain

$$\mu\{\sup_N |A_N(\boldsymbol{f}) - A_N(\boldsymbol{g})| > t\} \le \frac{(\sum_i \sqrt{e_i})^2}{t} \le \frac{3\sum_i e_i}{t}. \tag{9.5}$$

TR-item-168 also gives the valid $9t^{-2}\sum_i \|f_i - g_i\|_2^2$ bound. One may take the smaller of the two. Equation (9.5) is a direct bounded-input endpoint refinement, not a claim that one inequality dominates the other for every error and threshold. It gives another explicit Banach-principle closure once an actually dense bounded convergence class has been proved. The characteristic-factor space need not be dense in the whole space.

9.5 Finite oscillation perturbations

Let $Z_k, \widetilde{Z}_k$, $k_- \le k \le k_+$, be a true and an approximating finite schedule, with $\left\|Z_k - \widetilde{Z}_k\right\|_2 \le \epsilon_k$. For any fixed endpoint blocks, the difference between their integral oscillation sums is at most

$$\sum_{i=1}^{w} \left[\left(\sum_{k=k_i}^{k_{i+1}} \epsilon_k^2 \right)^{1/2} + \epsilon_{k_i} \right]. \tag{9.6}$$

Indeed, the difference of two maxima is bounded by the maximum of their pointwise differences; each increment error is bounded by $|Z_k - \widetilde{Z}_k| + |Z_{k_i} - \widetilde{Z}_{k_i}|$. Use the pointwise bound of a finite maximum by an ℓ^2 norm, then Cauchy–Schwarz in space. This proves (9.6) without a stochastic sampling premise. It turns an exact finite model into a certified finite-horizon inequality when genuine approximation errors are available. A uniform-in-horizon control and a justified limit are still necessary before applying Theorem 3.1.

10 What cannot be inferred

10.1 The product sequence is not automatically stationary

Even with an independent Bernoulli base, the sequence $F_n = f_1(T^n x) f_2(T^{2n} x) f_3(T^{3n} x)$ is not generally stationary in n. Let the coordinates X_j be independent fair $\{0, 1\}$ variables and take $f_1 = f_2 = f_3 = X_0$. Then $F_n = X_n X_{2n} X_{3n}$ and $\mathbb{E}F_n = 1/8$ for $n \geq 1$, but

$$\mathbb{E}(F_1 F_2) = 1/32, \qquad \mathbb{E}(F_3 F_4) = 1/64.$$

The first pair uses five distinct coordinates and the second six. Thus even lag-one pair laws depend on the initial index. Stationary concentration or a stationary covariance formula cannot be imposed on F_n solely because T preserves measure. The source's fixed-lag invariance calculation avoids this error.

10.2 Norm convergence and square energy are insufficient

A bounded sequence can have square-summable adjacent increments without converging: $a_k = \sin \log(k{+}2)$ has $|a_{k+1} - a_k| \leq 1/(k{+}2)$, yet subsequences tend to 1 and -1. This is the obstruction noted in TR-item-168. One must control block maxima, jumps, or full variation, not merely $\sum_k |a_{k+1} - a_k|^2$. Likewise L^2 convergence of (1.1) cannot alone supply the pointwise conclusion. Theorem 3.1 uses an additional uniform deterministic-block premise.

Ordinary mixing, even if accompanied by a theorem about mixing of all finite orders, does not in this report give summable convergence rates or pointwise maximal control. No consequence of the separately considered multiple-mixing manuscript is used to bypass O. The local quantitative routes here stand on their stated operator hypotheses.

10.3 Endpoint and approximation boundaries

There are three different endpoints. The analytic parameter $q > 2$ is fixed strictly above two, with constants allowed to deteriorate; the integrability endpoint $\sum_i 1/p_i = 1$ is reached by weak maximal control and density; and all-N passage is an index endpoint controlled by countably many profiles and finite grids. These cannot be conflated.

Finite matrix conditioning does not prove continuum completeness. Finite rational tests do not establish uniformity over all depth trees. A small leakage value does not prove exact graph membership. A moment-generating representation does not create a trilinear arithmetic estimate. Each of these distinctions is necessary for a correct applicability assessment.

11 Benefits, costs, and further research

The direct benefits are precise. The explicit moment bank gives an implementable profile construction with a matching leading lower bound. The row refinements reduce a linear delay cost to its square root

and a logarithmic-square cost to a logarithmic-3/2 cost. Symmetric planting improves a first-order scale error to a second-order one. The robust oscillation criterion quantifies a linear defect. Weak-mixing and L^p consequences show the genuine scope of the bounded analytic premise, including the endpoint L^1 output case.

Their costs are equally explicit. Moment cancellation pushes support outward. Persistent private coordinates require actual physical Gram control and can be expensive when their weights are small. Chronological row arguments require whole-cell supports and a common read per round. The L^p extension proves no new oscillation bound for unbounded inputs. The strongest general-system consequence depends on the unresolved verification of the source's universal analytic assertion.

These local improvements are applications of elementary Hilbert-space, moment, maximal, and convexity arguments. No claim of historical novelty for those mathematical tools is intended. The useful contribution is their explicit quantitative application with the original domains and quantifiers preserved.

A focused continuation of the mathematics would address:

1. An independent proof-level verification of the companion's sparse progression-test counting, all endpoint Bessel premises, and the absolute degree bounds used before q is selected.
2. A formal ledger separating original depths, accuracy levels, mode counts, and helper histories, with every estimate uniform in the original depth horizon.
3. Finite rational stress tests of whole-cell and shared-input hypotheses, including deliberately invalid fragmented-row variants, to detect misuse rather than to certify the infinite theorem.
4. A comparison of the explicit moment bank's support growth against the resulting profile-dependent C_φ, without assuming an effective mixing rate.
5. Further orthogonality savings within admissible row output spaces; square-summing is legitimate only when those output coordinates really are disjoint.

12 Evidence and reading boundaries

12.1 Local source locators

All paths below are local. The project root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`. The pinned M295 base directory is

```
_sources/openai-math/preprints/Pointwise-convergence-of-triple-ergodic-averages-for-mixing-transformations-October-4-2026/build/paper
```

The complete mathematical reading includes:

- `main.tex`, `preamble.tex`, `references.bib`, and `sections/introduction.tex`.
- `sections/ergodic.tex`: the whole section, especially `prop:orbit-oscillation`, `lem:ergodic-cauchy`, `eq:ergodic-finite-vdc`, `lem:ergodic-moments`, and the final proof of the main theorem.
- `sections/transfer.tex`: the whole section, including kernel construction, lattice translation, integer planting, prefix linearization, and orbit padding.
- `sections/rows.tex`: the whole section, especially `lem:row-maximal`, `lem:row-depth`, `cor:row-projections`, `eq:row-delay`, and `lem:smooth-depth`.
- `sections/base.tex`: the whole section, including capped fitting, cyclic accuracy, high replacements, delayed predictors, and active-cell reduction.
- `sections/localization.tex`: the whole section, including `eq:h-certified-variation`, `eq:h-frozen-composition`, stored side stacks, `eq:h-near-band`, and `eq:h-core-counts`.

- `sections/completion.tex`: the whole section, especially `eq:h-common-q`, `eq:h-uniform-fixed`, restarts, and finite accuracy-cutoff passage.
- `figures/row-chronology.tex` and `figures/frozen-depths.tex`.

The helper's JSON records SHA-256 hashes for the source TeX and bibliography files. A hash records identity, not verification.

The companion directory is the corresponding `An-L3-bound-for-the-trilinear-Hilbert-transform-October-5-2026/build`. The targeted reading comprised `sections/outer.tex` in full; `sections/compression.tex`, the nested-prefix lemma and Compression statement and proof; `sections/charts.tex`, Local detection's statement and its initial clipping and structured-pair argument; `sections/localization.tex`, Bessel separation of usable endpoint spaces and its proof; `sections/sides.tex`, Short-block comparison and its proof; and `sections/completion.tex`, Order of the parameters and Summing the base expansion. The entire companion chart, graph, mode, side, and core-counting construction was *not* independently audited. Its imported claims remain global verification gates. The earlier lemmas on which the selected companion arguments depend were not all re-proved.

The actual private sources, not only maps, were:

```
C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-168-a
stra.tex
C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/T
Rs/TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex
```

TR-item-168 was read in full, including its revised deterministic oscillation proof, fine-skeleton theorem, bounded approximation, and finite-horizon sections. For TR248, the reading covered Spaces and measurements; E1's Parseval, leakage, and finite-model proofs; Positive quadrature and physical coordinates; and Moment assembly, analytic tails, and exact certificates. No universal pointwise theorem is attributed to either corpus source.

12.2 Reproducible checks

The local helper `M295-Triple-Ergodic-Averages/_checks/check_m295.py` and its JSON output record **4,248 exact checks** and **298 floating-point diagnostics**, all passing. Exact checks include the formal four-direction coefficients, rational offset moments, translated-bump moment cancellation and total-variation bounds, the polynomial dual extrema, finite van der Corput inequalities, floor-grid coverage, a sharp delay multiplicity example, Bernoulli joint laws, finite cyclic approximation inequalities, and a positive physical Gram certificate. The floating checks test the displayed smooth-row numerical constant over a declared finite range; its general bound was proved above.

These tests are local algebra and finite inequalities. They do not verify an arbitrary adaptive continuum construction, the whole companion paper, the universal oscillation assertion, or a claimed convergence rate. The proofs, rather than sampling, justify the asymptotic and all-parameter results stated in this report.

An additional independent local review, recorded in `_audit/M295-source-audit.txt`, read all seven source sections and selected companion dependencies. It records 11,393 exact checks and 54 floating-point samples and agrees with the moment-bank and row refinements and the conditional general-system implication. It identified no definite fatal error within that reading scope, while explicitly leaving the companion's global analytic closure uncertified. Those separate counts are not included in the report helper's totals above.

12.3 Final assessment

The Ram corpus is directly applicable to M295's core operator and to specific substantial proof steps. The source-to-pointwise reduction, the explicit local refinements, and the conditional extensions have been distinguished throughout from the unverified global analytic closure. The report therefore preserves the high applicability rating without turning it into a success or validity rating. The conditional general-system consequence is an important logical clarification, not a claim that the longstanding problem has independently been solved.

References

[1] OpenAI, *Pointwise convergence of triple ergodic averages for mixing transformations*, October 4, 2026. New preprint; pinned repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned mathematical source. Source locators are specified in Section 12.

[2] OpenAI, *An L^3 bound for the trilinear Hilbert transform*, October 5, 2026. New companion preprint; same pinned repository commit. Pinned mathematical source. Sections 2–4, 6–7, 9–10 supply the imported statements discussed here.

[3] *Ram-Calculus Variation Certificates for Multiple Pointwise Ergodic Averages*, TR-item-168-astra, confidential local technical report, September 7, 2026, with subsequent correction and extension sections. Exact local file and reading scope in Section 12.

[4] M. Subbarao, *Stable Adaptive Ram Frames and Task-Aware Approximation*, TR248, confidential local corpus, 2026. Sections 2–4 and the moment-assembly appendix; exact local file in Section 12.

[5] B. Host and B. Kra, Nonconventional ergodic averages and nilmanifolds, *Annals of Mathematics* 161 (2005), 397–488. Theorem 1.1, norm convergence. `https://doi.org/10.4007/annals.2005.161.397`. Author-hosted theorem text: `https://sites.math.northwestern.edu/~kra/papers/convnil.pdf`.

[6] J. Bourgain, Double recurrence and almost sure convergence, *Journal für die reine und angewandte Mathematik* 404 (1990), 140–161. `https://doi.org/10.1515/crll.1990.404.140`. Its bilinear scope is also recorded in [10, Section 1.3, item 3]; the publisher full text was not available through the browsing interface.

[7] B. Krause, M. Mirek, and T. Tao, Pointwise ergodic theorems for non-conventional bilinear polynomial averages, *Annals of Mathematics* 195 (2022), 997–1109. `https://annals.math.princeton.edu/2022/195-3/p04`. Publisher theorem summary and DOI: `https://doi.org/10.4007/annals.2022.195.3.4`.

[8] W. Huang, S. Shao, and X. Ye, Pointwise convergence of multiple ergodic averages and strictly ergodic models, *Journal d'Analyse Mathématique* 139 (2019), 265–305. `https://doi.org/10.1007/s11854-019-0061-3`. The ergodic-distal conclusion is stated in the publisher abstract. Preprint: `https://arxiv.org/abs/1406.5930`.

[9] Y. Gutman, W. Huang, S. Shao, and X. Ye, Almost sure convergence of the multiple ergodic average for certain weakly mixing systems, arXiv:1612.02873v2, June 9, 2017. Main Theorem, p. 3, and Theorem 3.4: weak mixing plus PID. `https://arxiv.org/pdf/1612.02873v2`.

[10] D. Kosz, M. Mirek, S. Peluse, R. Wan, and J. Wright, The multilinear circle method and a question of Bergelson, arXiv:2411.09478v4, September 2, 2026 (PDF dated September 3). Theorems 1.3 and 1.12; Section 1.8, item 5, p. 22. https://arxiv.org/pdf/2411.09478v4. The record states that referees' reports have been incorporated; no additional publication status is inferred here.

[11] N. Frantzikinakis, *Progress since 2016 on some open problems on multiple ergodic averages*, latest stated update November 2025, Problem 11. https://users.math.uoc.gr/~frantzikinakis/OpenProblemsNew.html.